AN UPPER BOUND TO SECOND ORDER FOR THE LONGEST MINIMUM-WEIGHT PATH IN A COMPLETE GRAPH

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Abstract. Give the edges of the complete graph on n vertices independent exponential weights and let M_n be the largest number of edges of a minimum-weight path between two vertices. Addario-Berry, Broutin and Lugosi proved that $M_n/\log n \to \alpha$ in probability, where $\alpha \approx 3.5911$ solves $\alpha \log \alpha - \alpha = 1$, with the bounds $\alpha \log n - L \log\log n \le M_n \le \alpha \log n + \omega(n)$. We prove that $\mathbb{P}(M_n \ge b_n + x) \le C\alpha^{-x}$ for all $x \ge 0$, where $b_n = \alpha \log n - (\log\log n)/(2\log\alpha)$. Hence $M_n - \alpha \log n \to -\infty$ in probability and $\mathbb{E}M_n \le b_n + O(1)$. We also find the asymptotic value of the expected number of pairs of vertices joined by a minimum-weight path of at least k edges when $k = \alpha \log n + O(\log\log n)$, show that at $k = \lceil b_n \rceil$ the second moment of this number tends to infinity although its mean stays bounded, and prove that $\mathbb{P}(M_n = n-1) = n/(2\,(n-1)!)$ for every $n \ge 2$. The lower side of the question of van der Hofstad on the fluctuations of M_n stays open.

1. Introduction

Let K_n be the complete graph on the vertex set $[n] = \{1, \dots, n\}$ and give its edges independent weights with the exponential distribution of mean one. Almost surely no two distinct paths with the same endpoints have the same weight, so for $i \ne j$ there is a unique path of minimum weight from i to j; we write $H_n(i,j)$ for its number of edges, the *hopcount* between i and j. This is first-passage percolation on the complete graph, also called the stochastic mean-field model of distance. The weights of the minimum-weight paths were described by Janson [17]: divided by $\log n/n$, the weight between two fixed vertices, the largest weight from a fixed vertex and the largest weight over all pairs converge in probability to 1, 2 and 3; the fluctuations of the last quantity were found by Bhamidi and van der Hofstad [5]. Hopcounts for other edge-weight distributions on the complete graph are studied in [10, 11]. For exponential weights, the minimum-weight paths from a fixed vertex form a tree which, with its vertices listed in the order of their distance from the root, is a random recursive tree [14, 22]. Hence $H_n(1,2)$ is of order $\log n$, with Gaussian fluctuations of order $\sqrt{\log n}$ [14, 15], and

$$R_n = \max_j H_n(1,j)$$

has the law of the height of a random recursive tree with n vertices, and $R_n/\log n \to e$ in probability by theorems of Devroye [7] and Pittel [21].

The subject of this paper is the largest hopcount over all pairs,

$$M_n = \max_{1 \le i < j \le n} H_n(i,j).$$

Let $\alpha \approx 3.5911$ be the solution in (e, ∞) of

$$\alpha \log \alpha - \alpha = 1. \tag{1}$$

Janson [17] observed that the tail bound of Devroye for the height of a random recursive tree and a union bound over the n roots give $\mathbb{P}(M_n \ge \beta \log n) \to 0$ for every $\beta > \alpha$, and asked

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whether $M_n/\log n$ converges to a constant in $[e,\alpha]$. Addario-Berry, Broutin and Lugosi [1] proved that it converges to α. More precisely, their Theorem 1 states that for some constant L and every function $\omega(n)\to\infty$, with probability tending to one,

$$\alpha\log n - L\log\log n \le M_n \le \alpha\log n + \omega(n), \tag{2}$$

and that $\alpha\log n - L\log\log n \le \mathbb{E}M_n \le \alpha\log n + L$. In his Saint-Flour lectures, van der Hofstad [13, Open Problem 3.1] asks how the fluctuations of $R_n - e\log n$ and of $M_n - \alpha\log n$ behave. For the root the answer is known: Addario-Berry and Ford [2] proved that the height of a random recursive tree is $e\log n - \frac{3}{2}\log\log n + O(1)$ in expectation, with exponential tails around the expectation, and finer results on this height are in [19, 20, 6]. For M_n the bounds (2) are the best available.

Our main result removes the gap between the upper bound in (2) and the scale $\log\log n$ of the lower bound. Throughout, $\theta = \log\alpha$ and, for $n \ge 3$,

$$b_n = \alpha\log n - \frac{\log\log n}{2\theta}, \qquad \frac{1}{2\theta} \approx 0.3911. \tag{3}$$

Theorem 1. *There is a constant C such that for all $n \ge 3$ and all real $x \ge 0$,*

$$\mathbb{P}\big(M_n \ge b_n + x\big) \le C\,\alpha^{-x}.$$

Consequently $\mathbb{E}M_n \le b_n + O(1)$, and $\mathbb{P}(M_n \ge b_n + w_n) \to 0$ whenever $w_n \to \infty$. In particular $M_n - \alpha\log n \to -\infty$ in probability.

Together with the lower bound in (2), Theorem 1 shows that, with probability tending to one,

$$-L\log\log n \;\le\; M_n - \alpha\log n \;\le\; -\frac{\log\log n}{2\theta} + w_n$$

for every sequence $w_n \to \infty$.

The proof is a first-moment computation in which the factor $(\log n)^{-1/2}$ of the local limit theorem is kept. For $k \ge 1$ let

$$N_{n,k} = \#\big\{\{i,j\} :\ i \ne j,\ H_n(i,j) \ge k\big\}$$

be the number of pairs at hopcount at least k, so that $\{M_n \ge k\} = \{N_{n,k} \ge 1\}$. The mean of $N_{n,k}$ is given exactly by the expected profile of the random recursive tree (Lemma 5), and its asymptotic behaviour near b_n is as follows.

Theorem 2. *Let k_n be integers with $k_n - \alpha\log n = O(\log\log n)$. Then*

$$\mathbb{E}N_{n,k_n} = \big(C_* + o(1)\big)\,\alpha^{-(k_n - b_n)}, \qquad C_* = \frac{1}{2\,\Gamma(1+\alpha)\,(1-1/\alpha)\sqrt{2\pi\alpha}} \approx 0.01104.$$

Thus b_n is the threshold of the first moment: $\mathbb{E}N_{n,k}$ is of order one exactly when $k = b_n + O(1)$. Whether $M_n - b_n$ is tight is not known. For the root, the same computation puts the first-moment threshold of the height at $e\log n - \frac{1}{2}\log\log n$, whereas the expected height is $e\log n - \frac{3}{2}\log\log n + O(1)$ by [2], as for the maximum of a branching random walk. The level $\alpha\log n$, however, lies in the large deviation range of the height of a single tree, where the probability of reaching a level and the expected number of vertices at it differ at most by a power of $\log n$ [6], so the comparison with the root does not predict the centring of M_n. What a second moment argument meets is made precise by the next statement: at the first-moment threshold the pairs at large hopcount come in clusters with a common endpoint, and the second moment of their number is unbounded.

Proposition 3. *There is a constant $c > 0$ such that, with $k_n = \lceil b_n \rceil$ and for all large n,*

$$\mathbb{E}\big[N_{n,k_n}^2\big] \;\geq\; c\, n^{\,5-\alpha\log 4}\,(\log n)^{\frac{\log 4}{2\theta}-\frac{3}{2}}, \qquad 5-\alpha\log 4 \approx 0.0216 > 0.$$

The pairs $\{i,j\}$, $\{i,j'\}$ with a common endpoint alone contribute this much. In particular $(\mathbb{E}N_{n,k_n})^2/\mathbb{E}[N_{n,k_n}^2] \to 0$.

So the second moment method applied to $N_{n,k}$ itself gives nothing at $k = b_n + O(1)$; the lower bound in (2) is obtained in [1] by applying it to a restricted family of paths, which leads to the constant L. Proposition 3 does not decide whether $\mathbb{P}(M_n \geq b_n - A)$ is bounded away from zero for large A.

At the other end of the range the law of M_n can be written down.

Proposition 4. *For every $n \geq 2$,*

$$\mathbb{P}(M_n = n-1) = \frac{n}{2\,(n-1)!}.$$

For $n = 3, 4, 5, 6$ this gives $3/4$, $1/3$, $5/48$ and $1/40$.

The identity for $\mathbb{E}N_{n,k}$ and the tilting that leads to Theorems 1 and 2 are in Section 2, the two theorems are proved in Sections 3 and 4, Proposition 3 in Section 5 and Proposition 4 in Section 6, which also reports simulations of M_n.

2. The expected number of pairs at large hopcount

Fix a root $r \in [n]$. The union of the minimum-weight paths from r to the other vertices is a spanning tree, the shortest-path tree of r. Explore it by increasing distance from r. When m vertices have been found, the memoryless property of the exponential distribution makes the residual weights of the $m(n-m)$ edges between found and unfound vertices, that is, the weight of each such edge less the time elapsed since its found endpoint was reached, independent exponential variables of mean one, so the next vertex is attached by an edge chosen uniformly among them, and its parent is uniform among the m vertices found so far, independently of the past. The tree, with its vertices numbered in the order in which they are found, is therefore a random recursive tree: vertex $m+1$ is attached to a uniform vertex among $1, \dots, m$ [14, 22]. The hopcount $H_n(r,v)$ is the depth of v in this tree. The trees of different roots are dependent, and nothing below uses more than the law of one of them.

Let $Z_{n,d}$ be the number of vertices at depth d in a random recursive tree with n vertices, and

$$F_n(z) = \mathbb{E}\sum_{d\geq 0} Z_{n,d}\, z^d, \qquad f_{n,d} = [z^d]F_n(z) = \mathbb{E}Z_{n,d}.$$

A new vertex attached to a uniform vertex among n has depth one more than its parent, so $F_{n+1}(z) = (1+z/n)F_n(z)$ and

$$F_n(z) = \prod_{j=1}^{n-1}\Big(1+\frac{z}{j}\Big) = \frac{\Gamma(n+z)}{\Gamma(n)\,\Gamma(1+z)}. \tag{4}$$

Thus $(n-1)!\,f_{n,d}$ is the unsigned Stirling number of the first kind $\left[{n \atop d+1}\right]$; this description of the expected profile is classical, see [22], [9, Chapter 6] and [12].

Lemma 5. *For all $n \geq 2$ and $k \geq 1$,*

$$\mathbb{E}N_{n,k} = \frac{n}{2}\sum_{d\geq k} f_{n,d}.$$

Proof. Each pair $\{i,j\}$ is counted once from the root i and once from the root j, so $2N_{n,k} = \sum_r \#\{v \neq r : H_n(r,v) \geq k\}$. For each r the inner number is the number of vertices at depth at least k in the shortest-path tree of r, whose mean is $\sum_{d\geq k} f_{n,d}$. □

The sum in Lemma 5 is over the whole profile. It should not be replaced by $n-1$ times the tail of the hopcount between two fixed vertices computed from $F_n(z)/n$: that function is the generating function of the depth of a uniform vertex with the root included, and the depth of a uniform vertex other than the root has generating function $(F_n(z)-1)/(n-1)$.

Tilting. Let $a > 0$. Let $B_1, \dots, B_{n-1}$ be independent Bernoulli variables with

$$\mathbb{P}(B_j = 1) = q_j = \frac{a}{j+a}, \qquad S_n = \sum_{j=1}^{n-1} B_j.$$

Since $(1+au/j)/(1+a/j) = 1 - q_j + q_j u$, we have $\mathbb{E}u^{S_n} = F_n(au)/F_n(a)$ and therefore

$$f_{n,d} = F_n(a)\, a^{-d}\, \mathbb{P}(S_n = d) \qquad (d \geq 0). \tag{5}$$

For $a = 1$ this is the representation of the depth of a uniform vertex of the random recursive tree as a sum of independent indicators [8]. The mean and variance of S_n are

$$\mu_n = \sum_{j<n} \frac{a}{j+a}, \qquad V_n = \sum_{j<n} \frac{aj}{(j+a)^2}.$$

From $0 \leq a/j - a/(j+a) \leq a^2/j^2$ and $|aj/(j+a)^2 - a/j| \leq 2a^2/j^2 + a^3/j^3$,

$$\mu_n = a\log n + O(1), \qquad V_n = a\log n + O(1), \tag{6}$$

with constants depending on a only. As $V_n > 0$ for every $n \geq 2$, there is $c_a > 0$ with $V_n \geq c_a \log n$ for all $n \geq 2$.

Lemma 6. *Let X be a sum of finitely many independent Bernoulli variables, with variance $V > 0$. Then for all real t,*

$$\left|\mathbb{E}e^{itX}\right| \leq \exp\bigl(-2V\sin^2(t/2)\bigr), \tag{7}$$

and $\sup_d \mathbb{P}(X = d) \leq \sqrt{\pi/(8V)}$. In particular, for all $n \geq 2$,

$$\sup_d \mathbb{P}(S_n = d) \leq \sqrt{\frac{\pi}{8V_n}} \leq \frac{C_a}{\sqrt{\log n}}. \tag{8}$$

Proof. For one Bernoulli variable with parameter q, $|1-q+qe^{it}|^2 = 1 - 4q(1-q)\sin^2(t/2)$, and $\sqrt{1-y} \leq e^{-y/2}$ for $0 \leq y \leq 1$; multiplying over the summands gives (7). By Fourier inversion for an integer-valued variable,

$$\mathbb{P}(X=d) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \mathbb{E}e^{itX}\, e^{-itd}\, dt \leq \frac{1}{2\pi}\int_{-\pi}^{\pi} \left|\mathbb{E}e^{itX}\right| dt \leq \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-2Vt^2/\pi^2}\, dt = \sqrt{\frac{\pi}{8V}},$$

where we used $\sin(|t|/2) \geq |t|/\pi$ for $|t| \leq \pi$. □

3. Proof of Theorem 1

From now on $a = \alpha$ in the tilting of Section 2, and $\theta = \log \alpha$.

Lemma 7. *There is a constant C such that for all $n \geq 2$ and all integers $k \geq 1$,*

$$\mathbb{P}(M_n \geq k) \leq \mathbb{E}N_{n,k} \leq C\,\frac{n^{1+\alpha}\,\alpha^{-k}}{\sqrt{\log n}}.$$

Proof. The first inequality is Markov's. By Lemma 5, (5) and (8),

$$\mathbb{E}N_{n,k} = \frac{n}{2}F_n(\alpha)\sum_{d\geq k}\alpha^{-d}\,\mathbb{P}(S_n = d) \leq \frac{n}{2}F_n(\alpha)\,\frac{\alpha^{-k}}{1-1/\alpha}\,\frac{C_\alpha}{\sqrt{\log n}},$$

and $\log F_n(\alpha) \leq \alpha \sum_{j<n} 1/j \leq \alpha(1+\log n)$. □

Proof of Theorem 1. Let $n \geq 3$, write $\ell = \log n$, and let $x \geq 0$. Since M_n is an integer, $M_n \geq b_n + x$ exactly when $M_n \geq k$ with $k = \lceil b_n + x \rceil \geq 1$. By (1), $1 + \alpha - \alpha\theta = 0$, so

$$\frac{n^{1+\alpha}\,\alpha^{-b_n}}{\sqrt{\ell}} = \exp\Big((1+\alpha-\alpha\theta)\,\ell + \tfrac{1}{2}\log\ell\Big)\,\ell^{-1/2} = 1,$$

and Lemma 7 gives $\mathbb{P}(M_n \geq b_n + x) \leq C\alpha^{-(k-b_n)} \leq C\alpha^{-x}$. For the expectation, with $m = \lceil b_n \rceil$,

$$\mathbb{E}M_n \leq m + \sum_{j\geq 1}\mathbb{P}(M_n \geq m+j) \leq b_n + 1 + C\sum_{j\geq 1}\alpha^{-j}.$$

The remaining statements follow by taking $x = w_n$ and $x = (\log\log n)/(2\theta) - A$. □

Remark 8. The bound of [1] for the upper side of (2) uses the inequality $\mathbb{P}(\text{height} \geq x\log m) \leq e^{x-1}m^{x-x\log x}$, $x > 1$, of Devroye [7], which is the Chernoff bound for the sum in Lemma 5. Lemma 7 differs from it by the factor $(\log n)^{-1/2}$ of Lemma 6, and this factor is the source of the term $-(\log\log n)/(2\theta)$. For the height R_n of one tree and a fixed $\beta > e$, Chen and Ma [6, Proposition 3.1] prove $\mathbb{P}(R_n \geq \beta\log n) \leq Cn^{-\beta(\log\beta-1)}(\log n)^{-1/2}$, together with a lower bound of the same power of n. With $\beta = \alpha$ and a union bound over the roots their inequality gives $\mathbb{P}(M_n \geq \alpha\log n) = O((\log n)^{-1/2})$, which is the case $k = \lceil \alpha\log n\rceil$ of Lemma 7. The second-order term requires the bound at the levels $k = \alpha\log n - O(\log\log n)$, with a constant that does not depend on the level, and this is what Lemma 7 provides.

4. Proof of Theorem 2

We keep $a = \alpha$. The first step is a local limit theorem for sums of independent Bernoulli variables. It is standard; we include the short argument because it is used for S_n here and for two other families of sums in Section 5.

Lemma 9. *For $m \geq 1$ let X_m be a sum of finitely many independent Bernoulli variables, with mean $\mu^{(m)}$ and variance $V^{(m)}$, and let d_m be an integer. If $V^{(m)} \to \infty$ and $d_m - \mu^{(m)} = o\big(\sqrt{V^{(m)}}\big)$, then $\sqrt{V^{(m)}}\,\mathbb{P}(X_m = d_m) \to 1/\sqrt{2\pi}$.*

Proof. Let B be a Bernoulli variable with parameter q and $v = q(1-q)$. The centred variable $B - q$ has variance v and absolute third moment at most v, so for $|t| \leq 1$,

$$\mathbb{E}e^{it(B-q)} = 1 - \tfrac{1}{2}vt^2 + O(v|t|^3), \qquad \log\mathbb{E}e^{it(B-q)} = -\tfrac{1}{2}vt^2 + O\big(v|t|^3\big),$$

with absolute constants. Summing over the summands of X_m, we get $\log \mathbb{E}\exp(it(X_m - \mu^{(m)})) = -\frac{1}{2}V^{(m)}t^2 + O(V^{(m)}|t|^3)$ for $|t| \le 1$, and hence $\mathbb{E}\exp(iu(X_m - \mu^{(m)})/\sqrt{V^{(m)}}) \to e^{-u^2/2}$ for every real u. By Fourier inversion and the substitution $t = u/\sqrt{V^{(m)}}$,

$$\sqrt{V^{(m)}}\,\mathbb{P}(X_m = d_m) = \frac{1}{2\pi}\int_{-\pi\sqrt{V^{(m)}}}^{\pi\sqrt{V^{(m)}}} \mathbb{E}e^{iu(X_m-\mu^{(m)})/\sqrt{V^{(m)}}}\; e^{-iu(d_m-\mu^{(m)})/\sqrt{V^{(m)}}}\,du.$$

The integrand, set equal to zero outside the interval of integration, converges pointwise to $e^{-u^2/2}$ and, by (7) and $\sin(|t|/2) \ge |t|/\pi$, its modulus is at most e^{-2u^2/π^2}. Dominated convergence gives the limit $(2\pi)^{-1}\int e^{-u^2/2}du = 1/\sqrt{2\pi}$. □

Proof of Theorem 2. Write $\ell = \log n$. By (6), $\mu_n = \alpha\ell + O(1)$ and $V_n = \alpha\ell + O(1)$, so for each fixed integer $r \ge 0$ the point $k_n + r$ satisfies the hypothesis of Lemma 9 and

$$\sqrt{\ell}\,\mathbb{P}(S_n = k_n + r) \to \frac{1}{\sqrt{2\pi\alpha}}.$$

The left side is bounded uniformly in r and n by (8), and $\sum_r \alpha^{-r} < \infty$, so by dominated convergence

$$\sum_{r\ge 0}\alpha^{-r}\,\mathbb{P}(S_n = k_n + r) = \frac{1 + o(1)}{(1 - 1/\alpha)\sqrt{2\pi\alpha\,\ell}}.$$

By (4) and Stirling's formula, $F_n(\alpha) = (1 + o(1))\,n^\alpha/\Gamma(1 + \alpha)$. Lemma 5 and (5) now give

$$\mathbb{E}N_{n,k_n} = \frac{n}{2}F_n(\alpha)\,\alpha^{-k_n}\sum_{r\ge 0}\alpha^{-r}\,\mathbb{P}(S_n = k_n + r) = \frac{(1 + o(1))\;n^{1+\alpha}\,\alpha^{-k_n}}{2\,\Gamma(1 + \alpha)(1 - 1/\alpha)\sqrt{2\pi\alpha\,\ell}},$$

and $n^{1+\alpha}\alpha^{-b_n}/\sqrt{\ell} = 1$, as in the proof of Theorem 1. □

Remark 10. Theorem 2 is a statement about sums of Stirling numbers of the first kind in the range where the second index is close to α times the logarithm of the first. Asymptotic expansions of these numbers go back to Moser and Wyman [18]; the expansion of Hwang [16] is uniform for a second index up to any fixed multiple of the logarithm of the first, and Theorem 2 could also be derived from it.

Remark 11. The convergence in Theorem 2 is slow. For $a = \alpha$ one has $\mu_n = \alpha\log n - \alpha\psi(1 + \alpha) + o(1)$, where ψ is the digamma function and $\alpha\psi(1 + \alpha) \approx 5.07$, so at $k = b_n + O(1)$ the Gaussian factor $\exp(-(k - \mu_n)^2/(2V_n))$ of the local limit theorem differs from one by a term of order $(\log\log n)^2/\log n$, and for moderate n the constant $\alpha\psi(1 + \alpha)$ is the larger part of $k - \mu_n$. At $k = \lceil b_n\rceil$ the ratio of $\mathbb{E}N_{n,k}$, computed exactly from (4), to $C_*\alpha^{-(k-b_n)}$ is 0.388, 0.458, 0.614 and 0.718 for $n = 10^2, 10^3, 10^4, 10^5$; the Gaussian factor at the same points is 0.320, 0.436, 0.606 and 0.714.

5. The second moment at the first-moment threshold

Let $n \ge 4$ and $k \ge 1$, and let Y be the number of vertices at depth at least k in a random recursive tree with n vertices. Expanding the square of $N_{n,k}$ as a sum over two pairs, which coincide, share one vertex or are disjoint, and using the exchangeability of the vertices,

$$\mathbb{E}\big[N_{n,k}^2\big] = \binom{n}{2}p + 6\binom{n}{3}q_s + 6\binom{n}{4}q_d, \tag{9}$$

where $p = \mathbb{P}(H_n(1,2) \ge k)$, $q_s = \mathbb{P}(H_n(1,2) \ge k,\ H_n(1,3) \ge k)$ and $q_d = \mathbb{P}(H_n(1,2) \ge k,\ H_n(3,4) \ge k)$. The first two probabilities are determined by the shortest-path tree of vertex 1: the $n-1$ other vertices are exchangeable, so

$$p = \frac{\mathbb{E}Y}{n-1}, \qquad q_s = \frac{\mathbb{E}[Y(Y-1)]}{(n-1)(n-2)}, \qquad 6\binom{n}{3} q_s = n\,\mathbb{E}[Y(Y-1)].$$

The probability q_d involves two trees and is not determined by the law of one of them. We bound the middle term of (9) from below.

For a random recursive tree T_j with j vertices let

$$R_j(z,w) = \mathbb{E} \sum_{u \ne v} z^{\mathrm{depth}(u)}\, w^{\mathrm{depth}(v)},$$

the sum being over ordered pairs of distinct vertices of T_j.

Lemma 12. *For all $n \ge 2$,*

$$R_n(z,w) = \sum_{j=1}^{n-1} \frac{z+w}{j}\, F_j(zw) \prod_{l=j+1}^{n-1} \Big(1 + \frac{z+w}{l}\Big). \tag{10}$$

Proof. Write $P_j(z) = \sum_{u \in T_j} z^{\mathrm{depth}(u)}$. If vertex $j+1$ is attached to a vertex of depth D, the new ordered pairs contribute $z^{D+1}P_j(w) + w^{D+1}P_j(z)$. Given T_j, the parent is uniform, so the conditional mean of this is $j^{-1}(z+w)P_j(z)P_j(w)$, and $P_j(z)P_j(w)$ is the sum over ordered pairs of distinct vertices plus $P_j(zw)$. Hence

$$R_{j+1} = \Big(1 + \frac{z+w}{j}\Big) R_j + \frac{z+w}{j}\, F_j(zw), \qquad R_1 = 0,$$

and (10) follows by induction. □

In the jth term of (10) the factor $F_j(zw)$ accounts for the common part of the two paths from the root, which ends at the parent of vertex $j+1$, and the remaining factors account for the two separate parts. It is this term, for j a suitable power of n, that is large.

Proof of Proposition 3. Let $k = k_n = \lceil b_n \rceil$ and $T = \log n$. Write $Q_{j,n}(t) = \prod_{l=j+1}^{n-1}(1 + t/l)$ and $q_{j,h} = [t^h]Q_{j,n}(t)$. In the jth term of (10) the monomial $z^k w^k$ arises from the term $(zw)^d$ of $F_j(zw)$ and the term $z^s w^s$ of $(z+w)\,Q_{j,n}(z+w)$, where $s = k-d \ge 1$; the latter has coefficient $\binom{2s}{s} q_{j,2s-1}$. Therefore

$$[z^k w^k] R_n = \sum_{j=1}^{n-1} \frac{1}{j} \sum_{d<k} f_{j,d}\; q_{j,2(k-d)-1} \binom{2(k-d)}{k-d}. \tag{11}$$

All terms are nonnegative. Let D_j be the sum of independent Bernoulli variables with parameters $4/(l+4)$, $1 \le l < j$, and L_j the sum of independent Bernoulli variables with parameters $4/(l+4)$, $j < l < n$. As in (5),

$$f_{j,d} = F_j(4)\, 4^{-d}\, \mathbb{P}(D_j = d), \qquad q_{j,h} = Q_{j,n}(4)\, 4^{-h}\, \mathbb{P}(L_j = h),$$

and $F_j(4)\, Q_{j,n}(4) = F_n(4)/(1 + 4/j)$. With $s = k-d$, (11) becomes

$$[z^k w^k] R_n = F_n(4)\, 4^{1-k} \sum_{j=1}^{n-1} \frac{1}{j+4} \sum_{d<k} \mathbb{P}(D_j = d)\, \mathbb{P}(L_j = 2s-1) \binom{2s}{s} 4^{-s}. \tag{12}$$

Put $\rho = (\alpha-2)/2 \in (0,1)$ and $N = \lfloor n^\rho \rfloor$. We keep in (12) only the indices $N \le j \le 2N$ and, for each of them, the single value $d = d_j = \lfloor \mathbb{E}D_j \rfloor$. As in (6), uniformly in these j,

$$\mathbb{E}D_j = 4\log j + O(1) = 4\rho T + O(1), \qquad \mathbb{E}L_j = 4\log(n/j) + O(1) = 4(1-\rho)T + O(1),$$

and the variances of D_j and L_j satisfy the same estimates. Since $4\rho = 2\alpha - 4$ and $k_n = \alpha T + O(\log T)$, the number $s_j = k_n - d_j$ satisfies

$$s_j = (4-\alpha)T + O(\log T), \qquad 2s_j - 1 - \mathbb{E}L_j = O(\log T),$$

because $2(4-\alpha) = 4(1-\rho)$. Thus d_j and $2s_j - 1$ are within $o(\sqrt{T})$ of the means of D_j and L_j, whose variances are of order T. By Lemma 9, applied along an arbitrary sequence of pairs (n, j) with $n \to \infty$ and $N \le j \le 2N$, there is $c > 0$ such that for all large n and all such j,

$$\mathbb{P}(D_j = d_j) \ge \frac{c}{\sqrt{T}}, \qquad \mathbb{P}(L_j = 2s_j - 1) \ge \frac{c}{\sqrt{T}}.$$

Moreover $\binom{2s}{s} 4^{-s} \ge 1/(2\sqrt{s})$ for $s \ge 1$, so the last factor in (12) is at least $c/\sqrt{T}$; also $\sum_{j=N}^{2N} 1/(j+4) \ge 1/3$ for large N, and $F_n(4) = n(n+1)(n+2)(n+3)/24 \ge n^4/24$. Hence

$$[z^{k_n} w^{k_n}] R_n \;\ge\; c\, n^4\, 4^{-k_n}\, T^{-3/2}.$$

The left side is the expected number of ordered pairs of distinct vertices which are both at depth exactly k_n, and is at most $\mathbb{E}[Y(Y-1)]$. By (9) and the identities after it,

$$\mathbb{E}\big[N_{n,k_n}^2\big] \;\ge\; n\,\mathbb{E}[Y(Y-1)] \;\ge\; c\, n^5\, 4^{-k_n}\, T^{-3/2} \;\ge\; \frac{c}{4}\, n^{5-\alpha\log 4}\, T^{\frac{\log 4}{2\theta} - \frac{3}{2}},$$

where the last step uses $k_n \le b_n + 1$ and $4^{-b_n} = n^{-\alpha \log 4}\, T^{(\log 4)/(2\theta)}$. Since $\alpha < 3.5912$ and $\log 4 < 1.3863$, we have $\alpha \log 4 < 4.9785 < 5$. The right side therefore tends to infinity, while $\mathbb{E}N_{n,k_n}$ is bounded by Lemma 7. □

Remark 13. The proof shows which configurations are responsible. In the tree of a root i, the pairs kept in the proof are two vertices at depth k_n whose paths from i agree up to depth about $(2\alpha - 4)\log n \approx 3.18 \log n$ and then continue separately for about $(4-\alpha)\log n \approx 0.41 \log n$ edges each. The tree of a given root reaches depth k_n with probability at most $\mathbb{E}Y = O(1/n)$, by Lemma 7, whereas $\mathbb{E}[Y(Y-1)] \ge c\, n^{4-\alpha\log 4} (\log n)^{(\log 4)/(2\theta) - 3/2}$: a tree that reaches depth k_n carries on average at least of order $n^{5-\alpha\log 4}$ pairs of vertices at depth k_n, up to a power of $\log n$.

Remark 14. By the Cauchy–Schwarz inequality, $\mathbb{P}(M_n \ge k) \ge (\mathbb{E}N_{n,k})^2 / \mathbb{E}[N_{n,k}^2]$. Proposition 3 shows that this bound tends to zero at $k = \lceil b_n \rceil$, whatever the value of q_d. A second moment argument at this level would have to count a thinner family of paths, as in the truncated second moment arguments for the extremes of branching random walks [3, 4] and in [1, 2].

6. The top of the range, and simulations

Proof of Proposition 4. The coefficient of z^{n-1} in (4) is $1/(n-1)!$, so Lemma 5 gives $\mathbb{E}N_{n,n-1} = n/(2\,(n-1)!)$. It remains to show that $N_{n,n-1} \le 1$ almost surely. Suppose that the minimum-weight path between i and j has $n-1$ edges; it is then a Hamilton path γ. A subpath of a minimum-weight path is a minimum-weight path between its endpoints, since a lighter path between them would yield a lighter walk, and hence a lighter path, between i and j. By uniqueness, the minimum-weight path between any two vertices is the piece of γ between them, and it has $n-1$ edges only for the pair $\{i, j\}$. Therefore $N_{n,n-1}$ is the indicator of $\{M_n = n-1\}$. □

For $n = 3$ the proposition gives the law of M_3: it equals 2 with probability 3/4 and 1 with probability 1/4. In general $\mathbb{P}(M_n = n-1)$ is the probability that for some vertex the tree of minimum-weight paths from it is a path with that vertex at one end, and the proof shows that this then happens for exactly two vertices.

Finite checks and simulations. The identities of Sections 2 and 5 were checked by machine in small cases. The enumeration of all recursive trees with at most 8 vertices reproduces (4) and (11) in exact arithmetic, and for $K_3, \dots, K_6$ the values of $\mathbb{E}N_{n,k}$ in Lemma 5 and of $\mathbb{P}(M_n = n-1)$ agree with estimates from 10^5 sampled graphs within the sampling error (for $n = 5$, 0.1050 against $5/48 = 0.1042$).

Table 1 reports simulations of M_n for n between 250 and 4000. Each graph was sampled with independent exponential weights and the minimum-weight paths were found by Dijkstra's algorithm from every vertex; R_n denotes the largest hopcount from vertex 1, which is the height of a random recursive tree. For the comparison with the first moment let k_n^* be the real number at which $\mathbb{E}N_{n,k}$, computed exactly from (4) and interpolated linearly in $\log \mathbb{E}N_{n,k}$ between consecutive integers, equals one. By Theorem 2, $b_n - k_n^* \to \log(1/C_*)/\theta = 3.52$; for the values of n in the table it lies between 3.41 and 3.45.

n	graphs	mean M_n	s.e.	s.d.	range	mean R_n	b_n	k_n^*	mean $M_n - k_n^*$
250	8000	14.03	0.01	1.24	11–24	10.62	19.16	15.75	−1.72
500	4000	16.31	0.02	1.26	13–22	12.25	21.60	18.18	−1.86
1000	2000	18.66	0.03	1.31	15–25	13.90	24.05	20.61	−1.95
2000	400	20.95	0.06	1.27	18–25	15.74	26.50	23.06	−2.11
4000	120	23.36	0.10	1.13	21–28	17.13	28.96	25.50	−2.15

TABLE 1. Simulations of M_n and of R_n, the largest hopcount from one vertex.

n	k	$\mathbb{E}N_{n,k}$	freq.	k	$\mathbb{E}N_{n,k}$	freq.	k	$\mathbb{E}N_{n,k}$	freq.
250	16	0.7074	0.1128	17	0.1637	0.0352	18	0.0353	0.0083
500	19	0.3148	0.0505	20	0.0726	0.0143	21	0.0158	0.0030
1000	21	0.5897	0.0770	22	0.1429	0.0290	23	0.0329	0.0100
2000	24	0.2736	0.0325	25	0.0658	0.0075	26	0.0151	0.0000
4000	26	0.5133	0.0417	27	0.1279	0.0083	28	0.0306	0.0083

TABLE 2. The exact value of $\mathbb{E}N_{n,k}$ and the observed frequency of $\{M_n \geq k\}$ at the three integers k above k_n^*.

The mean of M_n lies between 1.7 and 2.2 below k_n^*, and so between 5.1 and 5.6 below b_n. The distance from the mean to b_n grows over the range of the table: it is larger at $n = 4000$ than at $n = 250$ by 0.47, with standard error 0.10, while $(\log\log n)/(2\theta)$ changes by 0.16. The standard deviation of M_n is between 1.1 and 1.3 for every n. Table 2 compares the frequency of $\{M_n \geq k\}$ with $\mathbb{E}N_{n,k}$, which bounds it from above. At the first integer above k_n^* the frequency is 0.16, 0.16, 0.13, 0.12 and 0.08 of $\mathbb{E}N_{n,k}$ for the five values of n, and at the other levels in the table it is at most 0.30 of it. A ratio well below one is to be expected: a minimum-weight path with $k+j$ edges contains $(j+1)(j+2)/2$ minimum-weight paths with at least k edges, so that the pairs counted by $N_{n,k}$ come in groups. The range of n is too small to show whether $M_n - b_n$ is tight.

One of the 8000 graphs with $n = 250$ has $M_n = 24$, while $\mathbb{P}(M_{250} \geq 24) \leq \mathbb{E}N_{250,24} = 9.3 \cdot 10^{-7}$, so that a sample of this size contains such a graph with probability at most 0.0075.

Its longest minimum-weight path was recomputed by the Bellman–Ford recursion over paths with a bounded number of edges (program `check_rep.c`): the path has 24 edges and weight 0.0415, and every path with at most 23 edges between its endpoints is heavier. No other graph with $n = 250$ has $M_n > 20$.

For a single root the comparison with the first moment can be made exactly. In a random recursive tree with $n \geq 2$ vertices the subtree of the second vertex has K vertices, where K is uniform on $\{1, \dots, n-1\}$ because the two sides of the tree grow as a Pólya urn started with one ball of each colour, and, given K, this subtree and the rest of the tree are independent random recursive trees with K and $n - K$ vertices. Hence $t_n(h) = \mathbb{P}(R_n > h)$ satisfies $t_n(-1) = 1$, $t_1(h) = 0$ for $h \geq 0$ and

$$t_n(h) = \frac{1}{n-1} \sum_{K=1}^{n-1} \Big(t_K(h-1)\big(1 - t_{n-K}(h)\big) + t_{n-K}(h) \Big), \qquad n \geq 2,\ h \geq 0.$$

The means of R_n computed from this recursion are 10.60, 12.25, 13.93, 15.63 and 17.34 for the five values of n in Table 1, and the simulated means differ from them by at most 1.5 standard errors. At $k = \lceil b_n \rceil$ the ratio of $\mathbb{P}(R_n \geq k)$ to the expected number $\sum_{d \geq k} f_{n,d}$ of vertices at depth at least k is 0.62, 0.56, 0.51, 0.49 and 0.49 for $n = 250$, 1000, 4000, 16000 and 32000.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [25] and Math Brain [23]. In October 2026 two AI agents connected to the Brain, working independently from the project's list of open problems in probability, obtained Theorems 1 and 2 and Propositions 3 and 4 with proofs, and a third agent checked the arguments. In November 2026 a Claude Fable agent (Anthropic) checked the proofs again, wrote and ran the programs and prepared the manuscript. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Programs and outputs:
`https://github.com/yspennstate/longest-geodesic-complete-graph`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [23].

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